STA 380

TUT 10

2. Consider the Anna-11 distribution with probability density function:

$$f_A(a) = \begin{cases} \frac{1}{(1-e^{-\rho})^{r-1}} \sum_{i=0}^{r-1} \binom{r-1}{i} (-1)^i e^{-\rho i} \boxed{\frac{1}{\Gamma(r-1)} (a-\rho i)^{(r-1)-1} e^{-(a-\rho i)}} \mathbf{1}_{\{a>\rho i\}}, & a \leq (r-1)\rho \\ 0 & \text{otherwise.} \end{cases}$$

Use `optimize()` to maximise the above pdf with respect to a between the interval $[0, 10]$.

Gamma pdf: $\frac{1}{\Gamma(\alpha) \cdot \theta^{\alpha}} x^{\alpha-1} \cdot e^{-\frac{x}{\theta}}$

Scale: 1 Shape: $r-1$ r.v: $x = a - \rho * i$

3. Consider the Laplace distribution with probability density function:

$$f(x \mid \mu, b) = \frac{1}{2b} \exp\left\{-\frac{|x-\mu|}{b}\right\}, \quad x \in \mathbb{R},\ \mu \in \mathbb{R},\ b > 0.$$

Here, μ is called the location parameter and b is called the diversity or scale parameter.

(a) Derive the log likelihood function.

① Likelihood

$$L(x \mid \mu, b) = \prod_{i=1}^{n} \left(\frac{1}{2b} \exp\left\{ -\frac{|x_i - \mu|}{b} \right\} \right)$$

$$= \left(\frac{1}{2b}\right)^n \prod_{i=1}^{n} \exp\left\{ -\frac{|x_i - \mu|}{b} \right\}$$

Review

$$e^a \cdot e^b = e^{a+b}$$

$$= \left(\frac{1}{2b}\right)^n \exp\left\{ -\frac{1}{b} \sum_{i=1}^{n} |x_i - \mu| \right\}$$

$\log(\exp\{\ldots\})$

② log-likelihood

$$\ell(x \mid \mu, b) = -n \log(2b) - \frac{1}{b} \sum_{i=1}^{n} |x_i - \mu|$$

$\log(2b^{-n})$

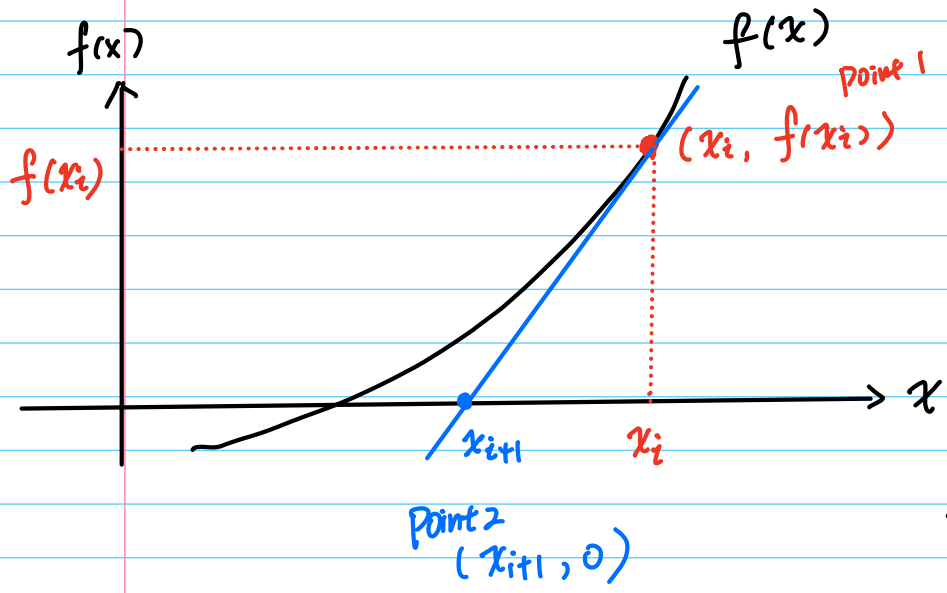
Newton - Raphson.

☆ : to find critical values

$f(x)$ $\qquad f'(x) = 0$

↓

log-likelihood function. $= \ell(\theta)$.



slope

↓

$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}}$$

$$\Rightarrow f'(x_i) = \frac{f(x_i)}{x_i - x_{i+1}}$$

$$\Rightarrow x_i - x_{i+1} = \frac{f(x_i)}{f'(x_i)}$$

$$\Rightarrow x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

↓

Optimization.

$$x_{i+1} = x_i - \frac{\ell'(\theta)}{\ell''(\theta)}$$

① Gradient. (first derivative)

⇒ direction of increase

② Hessian. (second derivative)

matrix

⇒ to find the curvature.

③ $\theta_{new} = \theta - H^{-1} \cdot \nabla \ell(\theta)$